

Multiple Regression Analysis: Estimation(1)

多元回归分析：估计（1）

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_k x_k + u$$

Chapter Outline 本章大纲

- **Motivation for Multiple Regression**

使用多元回归的动因

- **Mechanics and Interpretation of Ordinary Least Squares**

普通最小二乘法的操作和解释

- **The Expected Values of the OLS Estimators**

OLS估计量的期望值

- **The Variance of the OLS Estimators**

OLS估计量的方差

- **Efficiency of OLS: The Gauss-Markov Theorem**

OLS的有效性：高斯—马尔科夫定理

Lecture Outline 课堂大纲

- Motivation for multivariate Analysis 使用多元回归的动因
- The Model 模型
- The Estimation 估计
- Properties of the OLS estimates OLS估计的性质
- The Partialling out Interpretation 对“排除其它变量影响”的解释
- Simple versus multiple regressions 比较简单回归模型与多元回归模型
- Goodness of Fit 拟合优度

Motivation: Advantage

动因：优点

- The primary drawback of the simple regression analysis for empirical work is that it is very difficult to draw *ceteris paribus* conclusions about how x affects y .

在实证工作中使用简单回归模型的主要缺陷是：要得到在其它条件不变的情况下， x 对 y 的影响非常困难。

- Whether the *ceteris paribus* effects are reliable or not depends on whether the conditional mean assumption is realistic.

在其它条件不变情况假定下我们估计出的 x 对 y 的影响值是否可信依赖，完全取决于条件均值零值假设是否现实。

- If other factors that affecting y are not correlated with x , changing x can ensure that u is not changed, and the effect of x on y can be identified.

如果影响 y 的其它因素与 x 不相关，则改变 x 可以保证 u 不变，从而 x 对 y 的影响可以被识别出来。

Motivation : Advantage

动因：优点

- Multiple regression analysis is more amenable to ***ceteris paribus analysis*** because it allows us to explicitly control for many other factors that simultaneously affect the dependent variable.

多元回归分析更适合于其它条件不变情况下的分析，因为多元回归分析允许我们明确地控制许多其它也同时影响因变量的因素。

- Multiple regression models can accommodate many explanatory variables that may be correlated.

多元回归模型能容许很多解释变量，而这些变量可以是相关的。

- Important for drawing inference about causal relations between y and explanatory variables when using non-experimental data.

在使用非实验数据时，多元回归模型对推断 y 与解释变量间的因果关系很重要。

Motivation : Advantage

动因：优点

- It can explain more of the variation in the dependent variable.
它可以解释更多的因变量变动。
- It can incorporate more general functional form.
它可以表现更一般的函数形式。
- The multiple regression model is the most widely used vehicle for empirical analysis.
多元回归模型是实证分析中最广泛使用的工具。

Motivation: An Example

动因：一个例子

- Consider a simple version of the wage equation for obtaining the effect of education on hourly wage:
考虑一个简单版本的解释教育对小时工资影响的工资方程。

- *exper*: years of labor market experience
- *exper*: 在劳动力市场上的经历，用年衡量

$$wage = \beta_0 + \beta_1 educ + \beta_2 exper + u$$

- In this example experience (*exper*) is explicitly taken out of the error term.

在这个例子中，“在劳动力市场上的经历”被明确地从误差项中提出。

Motivation: An Example

动因：一个例子

- Consider a model that says family consumption is a quadratic function of family income:

考虑一个模型：家庭消费是家庭收入的二次方程。

$$\text{Cons} = \beta_0 + \beta_1 \text{inc} + \beta_2 \text{inc}^2 + u$$

- Now the marginal propensity to consume is approximated by

现在，边际消费倾向可以近似为

$$\text{MPC} = \beta_1 + 2\beta_2 \text{inc}$$

The Model with k Independent Variables

含有 k 个自变量的模型

- The general multiple linear regression model can be written as

一般的多元线性回归模型可以写为

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_k x_k + u$$

Parallels with Simple Regression

类似于简单回归模型

- β_0 is still the intercept β_0 仍是截距
- β_1 to β_k all called slope parameters β_1 到 β_k 都称为斜率参数
- u is still the error term (or disturbance) u 仍是误差项（或干扰项）
- Still need to make a zero conditional mean assumption, so now assume that 仍需作零条件期望的假设，所以现在假设

$$E(u|x_1, x_2, \dots, x_k) = 0$$

- Still minimizing the sum of squared residuals, so have $k+1$ first order conditions

仍然最小化残差平方和，所以得到**k+1**个一阶条件

Obtaining the OLS Estimates

如何得到OLS估计值

- The method of ordinary least squares chooses the estimates to minimize the sum of squared residuals,

普通最小二乘法选择能最小化残差平方和的估计值,

$$\sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \cdots - \hat{\beta}_k x_{ik})^2$$

Obtaining the OLS Estimates

如何得到OLS估计值

■ **k+1**个一阶条件为

$$\left\{ \begin{array}{l} \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \cdots - \hat{\beta}_k x_{ik}) = 0 \\ \sum_{i=1}^n x_{i1} (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \cdots - \hat{\beta}_k x_{ik}) = 0 \\ \sum_{i=1}^n x_{i2} (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \cdots - \hat{\beta}_k x_{ik}) = 0 \\ \cdots \\ \sum_{i=1}^n x_{ik} (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \cdots - \hat{\beta}_k x_{ik}) = 0 \end{array} \right.$$

Obtaining the OLS Estimates

如何得到OLS估计值

- The first order conditions are also the sample counterparts of the related population moments.
一阶条件也是相关的总体矩在样本中的对应。

- After estimation we obtain the OLS regression line, or the sample regression function (SRF)
在估计之后，我们得到OLS回归线，或称为样本回归方程（SRF）

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_{i1} + \dots + \hat{\beta}_k x_{ik}$$

Interpreting Multiple Regression

对多元回归的解释

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2 + \dots + \hat{\beta}_k x_k, \text{ so}$$

$$\Delta \hat{y} = \hat{\beta}_1 \Delta x_1 + \hat{\beta}_2 \Delta x_2 + \dots + \hat{\beta}_k \Delta x_k,$$

so holding $x_2, \dots, x_k$ fixed implies that

所以，保持 $x_2, \dots, x_k$ 不变意味着

$$\Delta \hat{y} = \hat{\beta}_1 \Delta x_1, \text{ that is each } \beta \text{ has}$$

a *ceteris paribus* interpretation

即，每一个 β 都有一个局部效应，或其它情况不变效应的意义。

Holding other factors fixed

“保持其它因素不变” 的含义

- The power of multiple regression analysis is that it allows us to do in non-experimental environments what natural scientists are able to do in a controlled laboratory setting: keep other factors fixed.

多元回归分析的优势在于它使我们能在非实验环境中去做自然科学家在受控实验中所能做的事情：保持其它因素不变。

Properties 性质

- The sample average of the residuals is zero.
残差项的样本平均值为零
- The sample covariance between each independent variable and the OLS residuals is zero.
每个自变量和OLS残差之间的样本协方差为零。
- The point $(\bar{x}_1, \bar{x}_2, \dots, \bar{x}_k, \bar{y})$ is always on the OLS regression line.
点 $(\bar{x}_1, \bar{x}_2, \dots, \bar{x}_k, \bar{y})$ 总位于OLS回归线上。

A “Partially Out” Interpretation

对“排除其它变量影响”的解释

- Consider regression line of 考虑回归线

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2$$

- One way to express $\hat{\beta}_1$ is $\hat{\beta}_1$ 的一种表达是

$$\hat{\beta}_1 = (\sum_{i=1}^n \hat{r}_{i1} y_i) / \sum_{i=1}^n \hat{r}_{i1}^2$$

- $\hat{r}_{i1}$ is obtained in the following way:

$\hat{r}_{i1}$ 由以下方式得出:

A “Partially Out” Interpretation

对“排除其它变量影响”的解释

- Regress our first independent variable x_1 on our second independent variable x_2 , and then obtain the residual r_1 . 将第一个自变量对第二个自变量进行回归，然后得到残差 r_1 。

- In other words, r_1 is the residual from the regression

$$\hat{x}_1 = \hat{\gamma}_0 + \hat{\gamma}_1 \hat{x}_2$$

换句话说， r_1 是由回归 $\hat{x}_1 = \hat{\gamma}_0 + \hat{\gamma}_1 \hat{x}_2$ 得到的残差。

- Then, do a simple regression of y on r_1 to obtain $\hat{\beta}_1$.

然后，将 y 向 r_1 进行简单回归得到 $\hat{\beta}_1$ 。

“Partialling Out” continued

“排除其它变量影响”（续）

- Previous equation implies that regressing y on x_1 and x_2 gives same effect of x_1 as regressing y on residuals from a regression of x_1 on x_2

上述方程意味着：将 y 同时对 x_1 和 x_2 回归得出的 x_1 的影响与先将 x_1 对 x_2 回归得到残差，再将 y 对此残差回归得到的 x_1 的影响相同。

- This means only the part of x_1 that is uncorrelated with x_2 are being related to y , so we're estimating the effect of x_1 on y after x_2 has been “partialled out”

这意味着只有 x_1 中与 x_2 不相关的部分与 y 有关，所以在 x_2 被“排除影响”之后，我们再估计 x_1 对 y 的影响。

“Partialling Out” continued

“排除其它变量影响”（续）

- In the general model with k explanatory variables, $\hat{\beta}_1$ can still be written as $\hat{\beta}_1 = (\sum_{i=1}^n \hat{r}_{i1} y_i) / \sum_{i=1}^n \hat{r}_{i1}^2$, but the residual r_{i1} comes from the regression of x_1 on $x_2 \dots, x_k$.

在一个含有 k 个解释变量的一般模型中， $\hat{\beta}_1$ 仍然可以写成 $\hat{\beta}_1 = (\sum_{i=1}^n \hat{r}_{i1} y_i) / \sum_{i=1}^n \hat{r}_{i1}^2$ ，但残差 r_{i1} 来自 x_1 对 $x_2 \dots, x_k$ 的回归。

- Thus $\hat{\beta}_1$ measures the effect of x_1 on y after $x_2, \dots, x_k$ has been partialled out.

于是 $\hat{\beta}_1$ 度量的是，在排除 $x_2 \dots, x_k$ 等变量的影响之后， x_1 对 y 的影响。

Simple vs Multiple Regression Estimates

比较简单回归和多元回归估计值

Compare the simple regression $\tilde{y} = \tilde{\beta}_0 + \tilde{\beta}_1 x_1$

比较简单回归 $\tilde{y} = \tilde{\beta}_0 + \tilde{\beta}_1 x_1$

with the multiple regression $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2$

与多元回归 $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2$

Generally, $\tilde{\beta}_1 \neq \hat{\beta}_1$ unless:

一般来说, $\tilde{\beta}_1 \neq \hat{\beta}_1$, 除非:

$\hat{\beta}_2 = 0$ (i.e. no partial effect of x_2) OR

$\hat{\beta}_2 = 0$ (也就是 x_2 对 y 没有局部效应), 或

x_1 and x_2 are uncorrelated in the sample

在样本中 x_1 和 x_2 不相关

Simple vs Multiple Regression Estimates

比较简单回归和多元回归估计值

- This is because there exists a simple relationship
这是因为存在一个简单的关系

$$\tilde{\beta}_1 = \hat{\beta}_1 + \hat{\beta}_2 \tilde{\delta}_1$$

- where $\tilde{\delta}_1$ is the slope coefficient from the simple regression of $\mathbf{x}_2$ on $\mathbf{x}_1$. The proof.

这里, $\tilde{\delta}_1$ 是 $\mathbf{x}_2$ 对 $\mathbf{x}_1$ 的简单回归得到的斜率系数。证明如下。

Because $y_i = \hat{\beta}_0 + \hat{\beta}_1 x_{i1} + \hat{\beta}_2 x_{i2} + \hat{u}_i$ so that

$y_i - \bar{y} = \hat{\beta}_1 (x_{i1} - \bar{x}_1) + \hat{\beta}_2 (x_{i2} - \bar{x}_2) + \hat{u}_i$, therefore

$$\tilde{\beta}_1 = \frac{\sum_{i=1}^n (x_{i1} - \bar{x}_1)(y_i - \bar{y})}{\sum_{i=1}^n (x_{i1} - \bar{x}_1)^2}$$

$$= \frac{\sum_{i=1}^n (x_{i1} - \bar{x}_1) [\hat{\beta}_1 (x_{i1} - \bar{x}_1) + \hat{\beta}_2 (x_{i2} - \bar{x}_2) + \hat{u}_i]}{\sum_{i=1}^n (x_{i1} - \bar{x}_1)^2}$$

$$= \hat{\beta}_1 + \hat{\beta}_2 \frac{\sum_{i=1}^n (x_{i1} - \bar{x}_1)(x_{i2} - \bar{x}_2)}{\sum_{i=1}^n (x_{i1} - \bar{x}_1)^2}$$

$$= \hat{\beta}_1 + \hat{\beta}_2 \tilde{\delta}_1$$

Simple vs Multiple Regression Estimates

简单回归和多元回归估计值的比较

Let $\hat{\beta}_j, j = 0, 1, \dots, k$ be the OLS estimators from the regression using full set of explanatory variables.

令 $\hat{\beta}_j, j = 0, 1, \dots, k$ 为用全部解释变量回归的OLS估计量。

Let $\tilde{\beta}_j, j = 0, 1, \dots, k-1$ be the OLS estimators from the regression that leaves out x_k .

令 $\tilde{\beta}_j, j = 0, 1, \dots, k-1$ 为用除 x_k 外的解释变量回归的OLS估计量。

Let $\tilde{\delta}_j$ be the slope coefficient on x_j in the regression of x_k on $x_1, \dots, x_{k-1}$. Then

令 $\tilde{\delta}_j$ 为 x_k 向 $x_1, \dots, x_{k-1}$ 回归中 x_j 的斜率系数。那么

$$\tilde{\beta}_j = \hat{\beta}_j + \hat{\beta}_k \tilde{\delta}_j.$$

Simple vs Multiple Regression Estimates

简单回归和多元回归估计值的比较

- In the case with k independent variables, the simple regression and the multiple regression produce identical estimate for x_1 only if

在 k 个自变量的情况下，简单回归和多元回归只有在以下条件下才能得到对 x_1 相同的估计

- (1) the OLS coefficients on x_2 through x_k are all zero, or

(1) 对从 x_2 到 x_k 的OLS系数都为零，或

- (2) x_1 is uncorrelated with each of $x_2 \dots, x_k$.

(2) x_1 与 $x_2 \dots, x_k$ 中的每一个都不相关。

Summary 总结

- In this lecture we introduce the multiple regression.

在本次课中，我们介绍了多元回归。

- Important concepts:

重要概念：

- Interpreting the meaning of OLS estimates in multiple regression

解释多元回归中OLS估计值的意义

- Partialling effect

局部效应（其它情况不变效应）

- Properties of OLS

OLS的性质

- When will the estimates from simple and multiple regression to be identical

什么时候简单回归和多元回归的估计值相同

Multiple Regression Analysis: Estimation (2)

多元回归分析：估计（2）

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_k x_k + u$$

Chapter Outline 本章大纲

- Motivation for Multiple Regression

使用多元回归的动因

- Mechanics and Interpretation of Ordinary Least Squares

普通最小二乘法的操作和解释

- **The Expected Values of the OLS Estimators**

OLS估计量的期望值

- **The Variance of the OLS Estimators**

OLS估计量的方差

- Efficiency of OLS: The Gauss-Markov Theorem

OLS的有效性：高斯—马尔科夫定理

Lecture Outline 课堂大纲

- The MLR.1 – MLR.4 Assumptions

假定MLR.1 – MLR.4

- The Unbiasedness of the OLS estimates

OLS估计值的无偏性

- Over or Under specification of models

模型设定过度或设定不足

- Omitted Variable Bias

遗漏变量的偏误

- Sampling Variance of the OLS slope estimates

OLS斜率估计量的抽样方差

The expected value of the OLS estimators

OLS估计量的期望值

- We now turn to the statistical properties of OLS for estimating the parameters in an underlying population model.

我们现在转向OLS的统计特性，而我们知道OLS是估计潜在的总体模型参数的。

- Statistical properties are the properties of estimators when random sampling is done repeatedly. We do not care about how an estimator does in a specific sample.

统计性质是估计量在随机抽样不断重复时的性质。我们并不关心在某一特定样本中估计量如何。

Assumption MLR.1 (Linear in Parameters)

假定 MLR.1（对参数而言为线性）

- In the population model (or the true model), the dependent variable y is related to the independent variable x and the error u as

在总体模型(或称真实模型)中, 因变量 y 与自变量 x 和误差项 u 关系如下

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_k x_k + u$$

where $\beta_1, \beta_2, \dots, \beta_k$ are the unknown parameters of interest, and u is an unobservable random error or random disturbance term.

其中, $\beta_1, \beta_2, \dots, \beta_k$ 为所关心的未知参数, u 为不可观测的随机误差项或随机干扰项。

Assumption MLR.2 (Random Sampling)

假定 MLR.2（随机抽样性）

- We can use a *random* sample of size n from the population,

我们可以使用总体的一个容量为 n 的随机样本 $\{(x_{i1}, x_{i2}, \dots, x_{ik}; y_i): i=1, \dots, n\}$,

where i denotes observation, and $j=1, \dots, k$ denotes the j th regressor.

其中 i 代表观察， $j=1, \dots, k$ 代表第 j 个回归元

- Sometimes we write 有时我们将模型写为

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \dots + \beta_k x_{ik} + u_i$$

Assumptions MLR.3 假定 MLR.3

- MLR.3 (Zero Conditional Mean) (零条件均值) :

$$E(u | x_{i1}, x_{i2}, \dots, x_{ik}) = 0.$$

When this assumption holds, we say all of the explanatory variables are *exogenous*; when it fails, we say that the explanatory variables are *endogenous*.

当该假定成立时，我们称所有解释变量均为*外生的*；否则，我们则称解释变量为*内生的*。

- We will pay particular attention to the case that assumption 3 fails because of omitted variables. 我们将特别注意当重要变量缺省时导致假定3不成立的情况。

Assumption MLR.4 假定MLR.4

- MLR.4 (No perfect collinearity) (不存在完全共线性) :
In the sample, none of the independent variables is constant, and there are no exact linear relationships among the independent variables.
- When one regressor is an exact linear combination of the other regressor(s), we say the model suffers from perfect collinearity.

在样本中，没有一个自变量是常数，自变量之间也不存在严格的线性关系。

当一个自变量是其它解释变量的严格线性组合时，我们说此模型有严格共线性。

Assumption MLR.4 假定MLR.4

- Example:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + u, \quad x_2 = 3x_3,$$

$$y = \beta_0 + \beta_1 \log(\text{inc}) + \beta_2 \log(\text{inc}^2) + u$$

$$\text{vote} = \beta_0 + \beta_1 \text{expendA} + \beta_2 \text{expendB} + \beta_3 \text{totexpend}_3 + u$$

- Perfect collinearity also happens when $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + u$, $n < (k+1)$.

当 $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + u$, $n < (k+1)$ 也发生完全共线性的情况。

- The denominator of the OLS estimator is 0 when there is perfect collinearity, hence the OLS estimator cannot be performed.

Theorem 3.1 (Unbiasedness of OLS)

定理 3.1 (OLS的无偏性)

- Under assumptions MLR.1 through MLR.4, the OLS estimators are unbiased estimator of the population parameters, that is
在假定MLR.1~MLR.4下，OLS估计量是总体参数的无偏估计量，即

$$E(\hat{\beta}_j | x_1, \dots, x_k) = \beta_j, \quad j = 1, 2, \dots, k.$$

Theorem 3.1 (Unbiasedness of OLS)

定理 3.1（OLS的无偏性）

- Unbiasedness is the property of an estimator, that is, the procedure that can produce an estimate for a specific sample, not an estimate. 无偏性是估计量的特性，而不是估计值的特性。估计量是一种方法（过程），该方法使得给定一个样本，我们可以得到一组估计值。我们评价的是方法的优劣。
- Not correct to say “5 percent is an unbiased estimate of the return of education”. 不正确的说法：“5%是教育回报率的无偏估计值。”

Too Many or Too Few Variables

变量太多还是太少了？

- What happens if we include variables in our specification that don't belong?

如果我们在设定中包含了不属于真实模型的变量会怎样？

- A model is *overspecified* when one or more of the independent variables is included in the model even though it has no partial effect on y in the population.

尽管一个（或多个）自变量在总体中对 y 没有局部效应，但却被放到了模型中，则此模型被*过度设定*。

- There is no effect on our parameter estimate, and OLS remains unbiased. But it can have undesirable effects on the variances of the OLS estimators.

过度设定对我们的参数估计没有影响，OLS仍然是无偏的。但它对OLS估计量的方差有不利影响。

Too Many or Too Few Variables

变量太多还是太少了？

- What if we exclude a variable from our specification that does belong?

如果我们在设定中排除了一个本属于真实模型的变量会如何？

- If a variable that actually belongs in the true model is omitted, we say the model is *underspecified*.

如果一个实际上属于真实模型的变量被遗漏，我们说此模型设定不足。

- OLS will usually be biased.

此时OLS通常有偏。

- Deriving the bias caused by omitting an important variable is an example of misspecification analysis.

推导由遗漏重要变量所造成的偏误，是模型设定分析的一个例子。

Omitted Variable Bias

遗漏变量的偏误

Suppose the true model is given as

假定真实模型如下

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + u,$$

but we estimate $\tilde{y} = \tilde{\beta}_0 + \tilde{\beta}_1 x_1 + v$, then

但我们估计的是 $\tilde{y} = \tilde{\beta}_0 + \tilde{\beta}_1 x_1 + v$, 有

$$\tilde{\beta}_1 = \frac{\sum (x_{i1} - \bar{x}_1) y_i}{\sum (x_{i1} - \bar{x}_1)^2}$$

Omitted Variable Bias Summary

遗漏变量的偏误 总结

- Two cases where bias is equal to zero 两种偏误为零的情形
 - $\beta_2 = 0$, that is x_2 doesn't really belong in model
 $\beta_2 = 0$, 也就是, x_2 实际上不属于模型
 - x_1 and x_2 are uncorrelated in the sample
样本中 x_1 与 x_2 不相关
- If correlation between x_2 , x_1 and x_2 , y is the same direction, bias will be positive 如果 x_2 与 x_1 间相关性和 x_2 与 y 间相关性同方向, 偏误为正。
- If correlation between x_2 , x_1 and x_2 , y is the opposite direction, bias will be negative 如果 x_2 与 x_1 间相关性和 x_2 与 y 间相关性反方向, 偏误为负。

Omitted Variable Bias Summary

遗漏变量的偏误 总结

When $E(\tilde{\beta}_1) > \beta_1$, we say that $\tilde{\beta}_1$ has upward bias.

当 $E(\tilde{\beta}_1) > \beta_1$, 我们说 $\tilde{\beta}_1$ 上偏。

When $E(\tilde{\beta}_1) < \beta_1$, we say that $\tilde{\beta}_1$ has downward bias.

当 $E(\tilde{\beta}_1) < \beta_1$, 我们说 $\tilde{\beta}_1$ 下偏。

Summary of Direction of Bias

偏误方向总结

$$\tilde{\beta}_1 = \hat{\beta}_1 + \hat{\beta}_2 \tilde{\delta}_1$$

	$\text{Corr}(x_1, x_2) > 0$	$\text{Corr}(x_1, x_2) < 0$
$\beta_2 > 0$	Positive bias 偏误为正	Negative bias 偏误为负
$\beta_2 < 0$	Negative bias 偏误为负	Positive bias 偏误为正

Omitted-Variable Bias 遗漏变量偏误

- In general , β_2 is unknown; and when a variable is omitted, it is mainly because of this variable is unobserved. In other words, we do not know the sign of $\text{Corr}(x_1, x_2)$. What to do? 但是，通常我们不能观测到 β_2 ，而且，当一个重要变量被缺省时，主要原因也是因为该变量无法观测，换句话说，我们无法准确知道 $\text{Corr}(x_1, x_2)$ 的符号。怎么办呢？
- We rely on economic theories and intuition to make an educated guess of the sign. 我们将依靠经济理论和直觉来帮助我们对相应符号做出较好的估计。

Example: hourly wage equation

例子：小时工资方程

- Suppose the model $\log(wage) = \beta_0 + \beta_1 educ + \beta_2 abil + u$ is estimated with *abil* omitted. What is the direction of bias for β_1 ?

假定模型 $\log(wage) = \beta_0 + \beta_1 educ + \beta_2 abil + u$ ，在估计时遗漏了 *abil*。 β_1 的偏误方向如何？

- Since in general ability has positive partial effect on y and ability and education years is positive correlated, we expect β_1 to have a *upward bias*.

因为一般来说ability对 y 有正的局部效应，并且ability和education years正相关，所以我们预期 β_1 上偏。

The More General Case

更一般的情形

- Technically, it is more difficult to derive the sign of omitted variable bias with multiple regressors.

从技术上讲，要推出多元回归下缺省一个变量时各个变量的偏误方向更加困难。

- But remember that if an omitted variable has partial effects on y and it is correlated with at least one of the regressors, then the OLS estimators of ***all*** coefficients will be biased.

我们需要记住，若有一个对 y 有局部效应的变量被缺省，且该变量至少和一个解释变量相关，那么*所有*系数的OLS估计量都有偏。

The More General Case

更一般的情形

$$y_{true} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + u$$

$$\hat{y}_{model1} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2 + \hat{\beta}_3 x_3$$

$$\tilde{y}_{model2} = \tilde{\beta}_0 + \tilde{\beta}_1 x_1 + \tilde{\beta}_2 x_2$$

Suppose $corr(x_1, x_3) = 0, corr(x_2, x_3) \neq 0$.

It is not difficult to believe that $\tilde{\beta}_2$ is a biased estimator of β_2 . Will $\tilde{\beta}_1$ be unbiased?

若 $corr(x_1, x_3) = 0, corr(x_2, x_3) \neq 0$ 。

很容易想到 $\tilde{\beta}_2$ 是 β_2 的一个有偏估计量。

而 $\tilde{\beta}_1$ 是有偏的吗？

The More General Case

更一般的情形

Yes. This is because if we regress x_3 on x_1 and x_2 ,
的确。这是因为如果我们将 x_3 向 x_1 和 x_2 回归,

$$\tilde{x}_3 = \tilde{\delta}_0 + \tilde{\delta}_1 x_1 + \tilde{\delta}_2 x_2$$

We have the following relations hold:

我们有如下关系成立:

$$\tilde{\beta}_1 = \hat{\beta}_1 + \hat{\beta}_3 \tilde{\delta}_1, \tilde{\beta}_2 = \hat{\beta}_2 + \hat{\beta}_3 \tilde{\delta}_2.$$

When $\text{corr}(x_1, x_2) \neq 0$, then $\tilde{\delta}_1 \neq 0$ even if
 $\text{corr}(x_1, x_3) = 0$. Therefore,

当 $\text{corr}(x_1, x_2) \neq 0$, 即使 $\text{corr}(x_1, x_3) = 0$, 也有 $\tilde{\delta}_1 \neq 0$ 。因此,

$\tilde{\beta}_1$ is a biased estimator of β_1 .

$\tilde{\beta}_1$ 是 β_1 的一个有偏估计量。

Variance of the OLS Estimators

OLS估计量的方差

- Now we know that the sampling distribution of our estimate is centered around the true parameter。 现在我们知道估计值的样本分布是以真实参数为中心的。
- Want to think about how spread out this distribution is 我们还想知道这一分布的分散状况。
- Much easier to think about this variance under an additional assumption, so 在一个新增假设下，度量这个方差就容易多了，有：

Assumption MLR.5 (Homoskedasticity)

假定MLR.5（同方差性）

- Assume Homoskedasticity: 同方差性假定:
$$\text{Var}(u|x_1, x_2, \dots, x_k) = \sigma^2 .$$
- Means that the variance in the error term, u , conditional on the explanatory variables, is the same for all combinations of outcomes of explanatory variables.
意思是，不管解释变量出现怎样的组合，误差项 u 的条件方差都是一样的。
- If the assumption fails, we say the model exhibits heteroskedasticity.
如果这个假定不成立，我们说模型存在异方差性。

Variance of OLS (cont)

OLS估计量的方差（续）

- Let $\mathbf{x}$ stand for $(x_1, x_2, \dots, x_k)$ 用 $\mathbf{x}$ 表示 $(x_1, x_2, \dots, x_k)$
- Assuming that $\text{Var}(u|\mathbf{x}) = \sigma^2$ also implies that $\text{Var}(y|\mathbf{x}) = \sigma^2$ 假定 $\text{Var}(u|\mathbf{x}) = \sigma^2$, 也就意味着 $\text{Var}(y|\mathbf{x}) = \sigma^2$
- Assumption MLR.1-5 are collectively known as the **Gauss-Markov assumptions**.
假定MLR.1-5共同被称为高斯—马尔科夫假定

Theorem 3.2 (Sampling Variances of the OLS Slope Estimators)

定理 3.2 (OLS斜率估计量的抽样方差)

Given the Gauss-Markov Assumptions

给定高斯—马尔科夫假定

$$\text{Var}(\hat{\beta}_j) = \frac{\sigma^2}{SST_j(1 - R_j^2)}, \text{ where}$$

$$SST_j = \sum (x_{ij} - \bar{x}_j)^2 \text{ and } R_j^2 \text{ is the } R^2$$

from regressing x_j on all other x 's

$$\text{其中, } SST_j = \sum (x_{ij} - \bar{x}_j)^2,$$

R_j^2 是 x_j 向所有其它 x 回归所得到的 R^2

Interpreting Theorem 3.2

对定理3.2的解释

- Theorem 3.2 shows that the variances of the estimated slope coefficients are influenced by three factors:

定理3.2显示：估计斜率系数的方差受到三个因素的影响：

- The error variance

误差项的方差

- The total sample variation

总的样本变异

- Linear relationships among the independent variables

解释变量之间的线性相关关系

Interpreting Theorem 3.2: The Error Variance

对定理3.2的解释（1）：误差项方差

- A larger σ^2 implies a larger variance for the OLS estimators.
更大的 σ^2 意味着更大的OLS估计量方差。
- A larger σ^2 means more noises in the equation.
更大的 σ^2 意味着方程中的“噪音”越多。
- This makes it more difficult to extract the exact partial effect of the regressor on the regressand. 这使得得到自变量对因变量的准确局部效应变得更加困难。
- Introducing more regressors can reduce the variance. But often this is not possible, neither is it desirable. 引入更多的解释变量可以减小方差。但这样做不仅不一定可能，而且也不一定总令人满意。
- σ^2 does not depend on sample size. σ^2 不依赖于样本大小

Interpreting Theorem 3.2: The total sample variation 对定理3.2的解释（2）：总的样本变异

- A larger SST_j implies a smaller variance for the estimators, and *vice versa*. 更大的 SST_j 意味着更小的估计量方差，反之亦然。
- Everything else being equal, more sample variation in x is always preferred. 其它条件不变情况下， x 的样本方差越大越好。
- One way to gain more sample variation is to increase the sample size. 增加样本方差的一种方法是增加样本容量。
- This components of parameter variance depends on the sample size. 参数方差的这一组成部分依赖于样本容量。

Interpreting Theorem 3.2: multicollinearity

对定理3.2的解释（3）：多重共线性

- A larger R_j^2 implies a larger variance for the estimators. 更大的 R_j^2 意味着更大的估计量方差。
- A large R_j^2 means other regressors can explain much of the variations in x_j . 如果 R_j^2 较大，就说明其它解释变量解释可以解释较大部分的该变量。
- When R_j^2 is very close to 1, x_j is highly correlated with other regressors, this is called **multicollinearity**. 当 R_j^2 非常接近1时， x_j 与其它解释变量高度相关，被称为**多重共线性**。
- Severe multicollinearity means the variance of the estimated parameter will be very large. 严重的多重共线性意味着被估计参数的方差将非常大。

Interpreting Theorem 3.2: multicollinearity 对定理3.2的解释（3）：多重共线性

- Multicollinearity is a data problem.
多重共线性是一个数据问题。
- Could be reduced by appropriately dropping certain variables, or collecting more data, etc. 可以通过适当地舍弃某些变量，或收集更多数据等方法来降低。
- Notice that a high degree of correlation between certain independent variables can be irrelevant as to how well we can estimate other parameters in the model. 注意：虽然某些自变量之间可能高度相关，但与模型中其它参数的估计程度无关。

Summary 总结

■ Important points of this lecture:

本堂课重要的几点：

■ Gauss-Markov assumptions

高斯-马尔科夫假定

■ What is consequence of overspecification and underspecification

模型过度设定和设定不足的后果

■ What is omitted-variable bias

遗漏变量偏差是什么

■ What are the three components of the variances of the estimated parameter and how they will affect the magnitude of the variances.

被估计参数方差的三个组成部分是什么，以及它们如何影响被估计参数方差的大小。

Multiple Regression Analysis: Estimation (3)

多元回归分析：估计（3）

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_k x_k + u$$

Chapter Outline 本章大纲

- Motivation for Multiple Regression

使用多元回归的动因

- Mechanics and Interpretation of Ordinary Least Squares

普通最小二乘法的操作和解释

- The Expected Values of the OLS Estimators

OLS估计量的期望

- The Variance of the OLS Estimators

OLS估计量的方差

- Efficiency of OLS: The Gauss-Markov Theorem

OLS的有效性：高斯—马尔科夫定理

Lecture Outline 课堂大纲

- The tradeoff of bias and variance in misspecified models 误设模型中偏误和方差间的替代关系
- Estimating the error variance 估计误差项方差
- The Gauss-Markov Theorem 高斯—马尔科夫定理
- Goodness of Fit 拟合优度
- Sample problems 例题

Variances in Misspecified Models

误设模型中的方差

- The tradeoff between bias and variance is important for considering whether to include an additional variable in the regression. 在考虑一个回归模型中是否该包括一个特定变量的决策中，偏差和方差之间的消长关系是重要的。
- Suppose the true model is $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + u$ then we have 假定真实模型是 $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + u$ ，我们有

$$Var(\hat{\beta}_1) = \frac{\sigma^2}{SST_1(1 - R_1^2)}$$

Variances in Misspecified Models

误设模型中的方差

- Consider the misspecified model

考虑误设模型是 $\tilde{y} = \tilde{\beta}_0 + \tilde{\beta}_1 x_1$

the estimated variance is

估计的方差是 $Var(\tilde{\beta}_1) = \frac{\sigma^2}{SST_1}$

- When x_1 and x_2 has zero correlation,

当 x_1 和 x_2 不相关时 $Var(\tilde{\beta}_1) = Var(\hat{\beta}_1)$

otherwise

否则

$$Var(\tilde{\beta}_1) < Var(\hat{\beta}_1)$$

Consequences of Dropping x_2

舍弃 x_2 的后果

	$R_1^2=0$	$R_1^2 \sim 0$
$\beta_2=0$	Both estimates of β_1 are unbiased, variances the same 两个对β_1的估计都是无偏的，方差相同	Both estimates of β_1 are unbiased, dropping x_2 results in smaller variance 两个对β_1的估计量都是无偏的，舍弃x_2使得方差更小
$\beta_2 \sim 0$	Dropping x_2 gives unbiased estimates of β_1, but its variance is the same as that from the full model. 舍弃x_2, β_1的估计量无偏，方差和从完整模型得到的估计相同	Dropping x_2 gives biased estimates of β_1, but its variance is smaller 舍弃x_2导致对β_1的估计量有偏，但其方差变小

Variances in Misspecified Models

误设模型中的方差

- If $\beta_2 \neq 0$, some econometricians prefer comparing the likely size of the bias due to omitting x_2 with the reduction in the variance.

如果 $\beta_2 \neq 0$ ，一些计量经济学家建议，将因漏掉 x_2 而导致的偏误的可能大小与方差的降低相比较以决定漏掉该变量是否重要。

- Nowadays including x_2 is often favored because the induced multicollinearity is less important as the sample size grows, but the omitted-variable bias does not necessarily follow any pattern.

现在，我们更喜欢包含 x_2 ，因为随着样本容量的扩大，增加 x_2 导致的多重共线性变得不那么重要，但舍弃 x_2 导致的遗漏变量误差却不一定有任何变化模式。

Estimating the Error Variance

估计误差项方差

- We wish to form an unbiased estimator of σ^2 .
我们希望构造一个 σ^2 的无偏估计量
- If we knew u , an unbiased estimator of σ^2 can be formed by calculating the sample average of the u^2 . 如果我们知道 u ，通过计算 u^2 的样本平均可以构造一个 σ^2 的无偏估计量
- We don't know what the error variance, σ^2 , is, because we don't observe the errors, u_i . 我们观察不到误差项 u_i ，所以我们不知道误差项方差 σ^2 。

Estimating the Error Variance

估计误差项方差

- What we observe are the residuals, $\hat{u}_i$
我们能观察到的是残差项 $\hat{u}_i$ 。
 - We can use the residuals to form an estimate of the error variance 我们可以用残差项构造一个误差项方差的估计
$$\hat{\sigma}^2 = \left(\sum \hat{u}_i^2 \right) / (n - k - 1) \equiv SSR / df$$
 - $df = n - (k + 1)$, or $df = n - k - 1$
 - df (i.e. degrees of freedom) is the (number of observations) – (number of estimated parameters)
- df（自由度），是观察点个数－被估参数个数

Estimating the Error Variance

估计误差项方差

- The division of $n-k-1$ comes from $E(\text{Sum of squared residuals}) = (n-k-1) \sigma^2$.

上式中除以 $n-k-1$ 是因为残差平方和的期望值是 $(n-k-1)\sigma^2$.

- Why degree of freedom is $n-k-1$?

为什么自由度是 $n-k-1$

- Because $k+1$ restrictions are imposed when deriving the OLS estimates. That is, given $n-k-1$ residuals, the remaining $k+1$ residuals are known, hence the degree of freedom is $n-k-1$.

因为推导OLS估计时，加入了 $k+1$ 个限制条件。也就是说，给定 $n-k-1$ 个残差，剩余的 $k+1$ 个残差是知道的，因此自由度是 $n-k-1$ 。

Estimating the Error Variance

估计误差项方差

- **Theorem 3.3** (Unbiased estimation of σ^2)

Under the Gauss-Markov Assumptions MLR.1-5, we have

$$E(\hat{\sigma}^2) = \sigma^2$$

定理3.3（ σ^2 的无偏估计）在高斯—马尔科夫假定 MLR.1-5下，我们有 $E(\hat{\sigma}^2) = \sigma^2$ 。

- Terminology: The positive square root of σ^2 is called *standard deviation*, and the positive square of $\hat{\sigma}^2$ is called *standard error*.

σ^2 正的平方根称为 标准偏差（标准离差）（SD），

$\hat{\sigma}^2$ 正的平方根称为 标准误差（标准差）（SE）。

- The standard error of $\hat{\beta}_j$ is $se(\hat{\beta}_j) = \hat{\sigma} / [SST_j (1 - R_j^2)]^{1/2}$.

Efficiency of OLS: The Gauss-Markov Theorem

OLS的有效性：高斯—马尔科夫定理

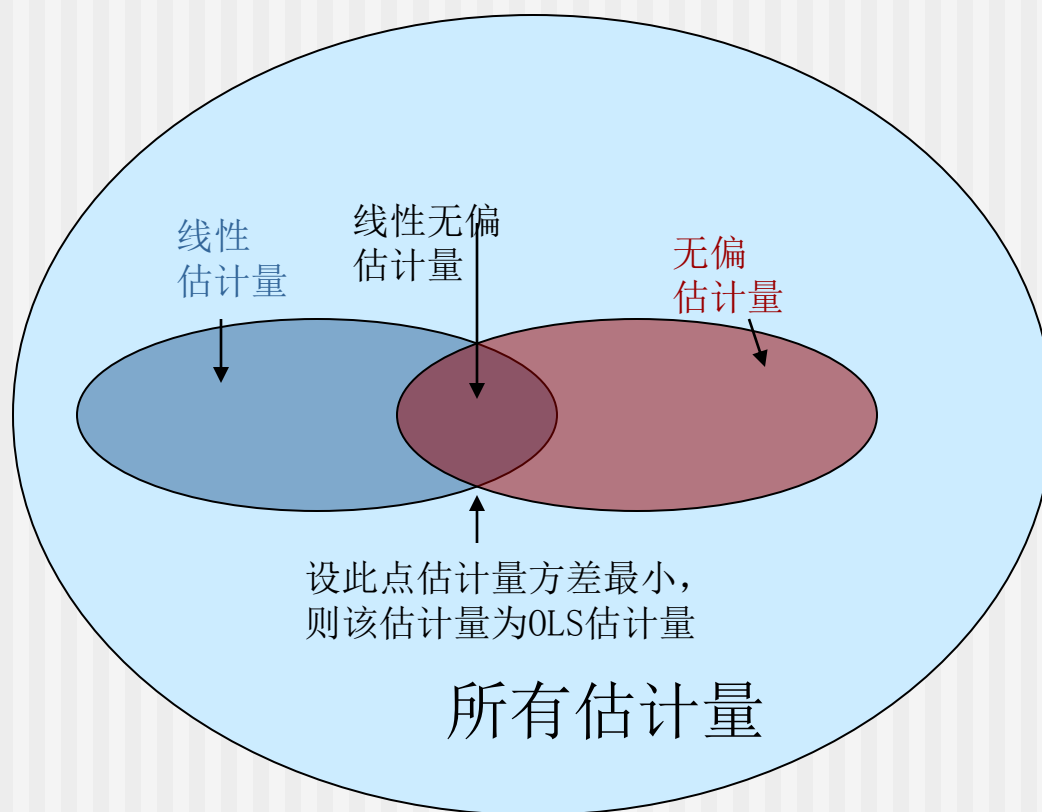
- Question: There are many unbiased estimators of β_j under MLR.1 – 5. Why OLS?

问题：在假定 MLR.1.5 下有许多 β_j 的估计量，为什么选 OLS？

- **OLS is Best Linear Unbiased Estimator (BLUE) under MLR.1 – 5.** 在假定 MLR.1.5 下，OLS 是最优线性无偏估计量（BLUE）。

- Best: smallest variance 最优：方差最小
- Linear: linear function of the data on the dependent variable
线性：因变量数据的线性函数
- Unbiased: the expectation of the estimated parameter equals its true value. 无偏：参数估计量的期望等于参数的真值。
- Estimator: a rule to produce an estimator. 估计量：产生一个估计量的规则

高斯—马尔科夫定理图示



The Importance of Gauss-Markov Theorem

高斯—马尔科夫定理的重要性

- When standard assumption holds, we need not look for alternative unbiased estimators.

当标准假定成立，我们不需要再去找其它无偏估计量了。

- If we are presented with an estimator that is both linear and unbiased, then we know that the variance of this estimator is at least as large as that from OLS.

如果有人向我们提出一个线性无偏估计量，那我们就知道，此估计量的方差至少和OLS估计量的方差一样大。

Some Details about Linearity of the OLS Estimator OLS估计量为线性的一些细节

以 $\hat{y} = \beta_0 + \hat{\beta}_1 x_1$ 为例，则

$$\hat{\beta}_1 = \frac{\sum (x_{i1} - \bar{x}_1) y_i}{\sum (x_{i1} - \bar{x}_1)^2} = \sum \frac{(x_{i1} - \bar{x}_1)}{\sum (x_{i1} - \bar{x}_1)^2} y_i.$$

$$\text{令 } w_i = \frac{(x_{i1} - \bar{x}_1)}{\sum (x_{i1} - \bar{x}_1)^2} \text{ 则 } \hat{\beta}_1 = \sum w_i y_i.$$

即 $\hat{\beta}_1$ 是相对于 y 的线性估计量。

Some Details about Linearity of the OLS Estimator

OLS估计量为线性的一些细节

w_i 有以下特性(请课后自己证明)：

$$\sum w_i = 0,$$

$$\sum w_i^2 = \frac{1}{\sum (x_{i1} - \bar{x}_1)^2}.$$

$$\sum w_i (x_{i1} - \bar{x}_1) = \sum w_i x_{i1} = 1.$$

Goodness-of-Fit 拟合优度

We can think of each observation as being made up of an explained part, and an unexplained part, 每一个观察值可被视为由解释部分和未解释部分构成, $y_i = \hat{y}_i + \hat{u}_i$.

Define: 定义:

$\sum (y_i - \bar{y})^2$: total sum of squares (SST) 总平方和

$\sum (\hat{y}_i - \bar{y})^2$: explained sum of squares (SSE) 解释平方和

$\sum \hat{u}_i^2$: residual sum of squares (SSR) 残差平方和

Then $SST = SSE + SSR$

有, $SST = SSE + SSR$

Goodness-of-Fit (continued)

拟合优度（续）

- ◆ How do we think about how well our sample regression line fits our sample data?

我们怎样衡量我们的样本回归线拟合样本数据有多好呢？

- ◆ Can compute the fraction of the total sum of squares (SST) that is explained by the model, call this the R-squared of regression

可以计算总平方和（SST）中被模型解释的部分，称此为回归 R^2

- ◆ $R^2 = SSE/SST = 1 - SSR/SST$

Goodness-of-Fit (continued)

拟合优度（续）

We can also think of R^2 as being equal to the squared correlation coefficient between the actual y_i and the values $\hat{y}_i$

我们也可以认为 R^2 等于实际的 y_i 与估计的 $\hat{y}_i$ 之间相关系数的平方

$$R^2 = \frac{\left(\sum (y_i - \bar{y})(\hat{y}_i - \bar{\hat{y}}) \right)^2}{\left(\sum (y_i - \bar{y})^2 \right) \left(\sum (\hat{y}_i - \bar{\hat{y}})^2 \right)}$$

More about R -squared

更多关于 R^2

- R^2 generally increases when a regressor is added to a regression.

当回归中加入另外的解释变量时， R^2 通常会上升。

- Exception: if the new regressor is perfectly multicollinear with the original regressors, then OLS cannot be implemented.

例外：如果这个新解释变量与原有的解释变量完全共线，那么OLS不能使用。

- This algebraic fact follows because the sum of squared residuals never increase when additional regressor are added to the model.

此代数事实成立，因为当模型加入更多回归元时，残差平方和绝不会增加。

More about R -squared

更多关于 R^2

- Think about starting with one regressor and then adding a second.

考虑从一个解释变量开始，然后加入第二个。

- Properties of OLS: minimize the sum of squared residuals.

OLS性质：最小化残差平方和。

- If OLS happens to choose the coefficient on the new regressor to be exactly zero, then SSR will be the same whether or not the second variable is included in the regression.

如果OLS恰好使第二个解释变量系数取零，那么不管回归是否加入此解释变量，SSR相同。

More about R -squared

更多关于 R^2

- If OLS chooses any value other than zero, it must be that this value reduced the SSR relative to the regression that excludes the regressor.

如果OLS使此解释变量取任何非零系数，那么加入此变量之后，SSR降低了。

- In practice it is extremely unusual for an estimated coefficient to be exactly zero, so in general the SSR will decrease when a new regressor is added.

实际操作中，被估计系数精确取零是极其罕见的，所以，当加入一个新解释变量后，一般来说，SSR会降低。

The Adjusted R^2

调整过的 R^2

- Therefore, an increase in the R^2 does not mean that adding a variable necessarily improves the fit of the model. 因此， R^2 增加并不意味着加入新的变量一定会提高模型拟合度。
- The adjusted R^2 is a modified version of the R^2 that does not necessarily increase when a new regressor is added.

调整过的 R^2 是 R^2 一个修正版本，当加入新的解释变量，调整过的 R^2 不一定增加。

$$\bar{R}^2 = 1 - \frac{SSR / (n - (k + 1))}{SST / (n - 1)} = 1 - \frac{n - 1}{n - k - 1} \frac{SSR}{SST}$$

The Adjusted R^2

调整过的 R^2

- The adjusted R^2 is 1 minus the ratio of sample variance of the OLS residuals (after correcting the degrees of freedom) to the sample variance of y .

调整过的 R^2 是1减去OLS残差的样本方差（修正过自由度之后）与 y 的样本方差之比。

- Three useful properties of adjusted R^2 :

调整过的 R^2 的三个有用性质：

- ① Since $(n-1)/(n-k-1) > 1$, the adjusted R^2 is always smaller than R^2 .

因为 $(n-1)/(n-k-1) > 1$ ，所以调整过的 R^2 总比 R^2 小。

The Adjusted R^2

调整过的 R^2

- ② Adding a regressor has two opposite effects.
(1) SSR falls so that adjusted R^2 should increase. (2) $(n-1)/(n-k-1)$ increase so that adjusted R^2 should decrease.

加入一个解释变量有两个相反的效果。(1) SSR降低导致调整过的 R^2 增加。(2) $(n-1)/(n-k-1)$ 增加导致调整过的 R^2 降低。

The Adjusted R^2 调整过的 R^2

- ③ The adjusted R^2 can be negative. This happens when the regressors, taken together, reduce the sum of squared residuals by such a small amount that this reduction fails to offset the factor $(n-1)/(n-k-1)$.

调整过的 R^2 可能是负的，发生在以下情况：所有解释变量使残差平方和下降的太少，不足以抵消因子 $(n-1)/(n-k-1)$ 。

- R^2 can be negative only in the case of regression through origin.

R^2 只有在过原点回归中才可能为负。

R^2 versus Adjusted R^2

比较 R^2 和 Adjusted R^2

- The R^2 and Adjusted R^2 tell us whether the regressors are good at predicting, or “explaining” the values of the dependent variable in the sample of data on hand.

R^2 和调整过的 R^2 告诉我们，解释变量是否很好地预测了，或“解释”了，手头数据中被解释变量的值。

R^2 versus Adjusted R^2

比较 R^2 和Adjusted R^2

- The R^2 and Adjusted R^2 do not tell us whether R^2 和调整过的 R^2 并没有告诉我们
 - An included variable is statistically significant
被包含变量是否统计显著
 - The regressors are a true cause of the movements in the dependent variable
解释变量是否是被解释变量变动的真正原因
 - There is omitted variable bias, or
是否有遗漏变量偏误，或
 - You have chosen the most appropriate set of regressors. 是否选取了最合适的解释变量组合

R^2 和Adjusted R^2

- Both R^2 and Adjusted R^2 are not good tools for deciding whether one variable should be added to a model.

在决定某个变量是否应该被加入模型时， R^2 和adjusted R^2 并非理想的工具。

- The factor that should determine whether an explanatory variable belongs in a model is whether the explanatory variable has a nonzero partial effect on y in the population.

决定一个解释变量是否属于模型的因素应该是，该解释变量在总体中对 y 的局部效应是否为零。

Review 复习

- Properties of the OLS estimators in multiple regression. 多元回归中OLS估计量的性质
- The Gauss-Markov assumptions and unbiasedness of the OLS estimators 高斯-马尔科夫假定和OLS估计量的无偏性
- How to calculate degree of freedom 如何计算自由度
- What are model over-specification, and under-specification, the tradeoff of expectation and variances in these two cases 模型过度设定和设定不足是什么，两种情况下，期望和方差间的替代关系
- What is omitted-variable bias, when this bias will be zero, how to determine the signs of this bias 遗漏变量偏误是什么，什么情况下此偏误为零，如何确定偏误符号

Review 复习

- What determines the variances of the OLS slope estimators, how to calculate standard deviation and standard error for them. How to estimate error variances, and to derive the standard deviation of the estimated parameters OLS斜率估计量方差由什么决定，如何计算它们的标准离差和标准差，如何估计误差项方差，以及如何推导被估参量的标准离差
- The additional assumption and the Gauss-Markov Theorem 新加的假定和高斯—马尔科夫定理
- R^2 和Adjusted R^2 R^2 和调整过的 R^2